

1. What is the use of 8051 chip?

Intel's 8251A is a universal synchronous asynchronous receiver and transmitter compatible with Intel's Processors. This may be programmed to operate in any of the serial communication modes built into it. This chip converts the parallel data into a serial stream of bits suitable for serial transmission. It is also able to receive a serial stream of bits and converts it into parallel data bytes to be read by a microprocessor.

2. What are the different types of methods used for data transmission?

The data transmission between points involves unidirectional or bi-directional transmission of meaningful digital data through a medium. There are basically three modes of data transmission

- ☐ Simplex
- ☐ Duplex
- ☐ Half Duplex

In simplex mode, data is transmitted only in one direction over a single communication channel. For example, a computer (CPU) may transmit data for a CRT display unit in this mode.

In duplex mode, data may be transferred between two trans receivers in both directions simultaneously.

In half duplex mode, on the other hand, data transmission may take place in either direction, but at a time may be transmitted only in one direction. For example, a computer may communication with a terminal in this mode. When the terminal sends data (i.e. terminal is sender). The message is received by the computer (i.e. computer is receiver). However, it is not possible to transmit data from the computer to terminal and from terminal to the computer simultaneously.

3. What is the various programmed data transfer method?

- ☐ Synchronous data transfer
 - ☐ Asynchronous data transfer
 - ☐ Interrupt driven data transfer

4. What is synchronous data transfer?

It is a data method which is used when the I/O device and the microprocessor match in speed. The transfer a data to or from the device, the user program issues a suitable instruction addressing the device. The data transfer is completed at the end of the execution of this instruction.

5. What is asynchronous data transfer?

It is a data transfer method which is used when the speed of I/O device does not match with the speed of the microprocessor. Asynchronous data transfer is also called as Handshaking.

6. What are the functional types used in control words of 8251a?

The control words of 8251A are divided into two functional types

Mode Instruction control word: - This defines the general operational characteristics of 8251A.
Command Instruction control word: - The command instruction controls the actual operations of the selected format like enable transmit/receiver, error reset and modem control.

7. What are the basic modes of operation of 8255?

There are two basic modes of operation of 8255, viz.

- ☐ I/O mode.

- ☐ BSR mode

In I/O mode, the 8255 ports work as programmable I/O ports, while in BSR mode only port C (PC0-PC7) can be used to set or reset its individual port bits. Under the IO mode of operation, further there are three modes of operation of 8255, So as to support different types of applications, viz. mode 0, mode 1, and mode 2.

Mode 0- Basic I/O mode

Mode 1-Strobe I/O mode

Mode 2- Strobe bi-direction I/O

8. Write the features of mode 0 in 8255?

- ☐☐ Two 8-bit ports (port A and port B) and two 4-bit ports (port C upper and lower) are available.
- ☐ The two 4-bit ports can be combined used as a third 8-bit port.

- ☐ Any port can be used as an input or output port.

- ☐ Output ports are latched. Input ports are not latched.

- ☐☐ A maximum of four ports are available so that overall 16 I/O configurations are possible.

9. What are the features used mode 1 in 8255?

- ☐ Two groups A and group B are available for strobe data transfer.

- ☐ Each group contains one 8-bit data I/O port and one 4-bit control/data port.

- ☐ The 8-bit data port can be either used as input or output port. The inputs and outputs both are latched.

- ☐ Out of 8-bit port C, PC0-PC2 is used to generate control signals for port B and PC3=PC5 are used to generate control signals for port A. The inputs PC6, PC7 may be used as independent data lines.

10. What are the signals used in input control signal and output control signals?

- ☐ Input control signals

- STB (Strobe input) IBF (Input buffer full) INTR (Interrupt request)

- ☐☐ Output control signal

- OBF (Output buffer full) ACK (Acknowledge input)

INTR (Interrupt request)

11. What are the features used mode 2 in 8255?

The signals 8-bit port in group A is available.

- ☐ The 8-bit port is bi-directional and additionally a 5-bit control port is available.
- ☐ Three I/O lines are available at port C, viz PC2-PC0.
 - ☐ Inputs and output are both latched.
 - ☐ The 5-bit control port C (PC3-PC7) is used for generating/accepting handshake Signals for the 8-bit data transfer on port A.

12. What are the modes of operation used in 8253?

Each of the three counters of 8253 can be operated in one of the following six modes of operation.

- ☐ Mode 0 (Interrupt on terminal count)
 - ☐ Mode 1 (Programmable monoshot)
 - ☐ Mode 2 (Rate generator)
 - ☐ Mode 3 (Square wave generator)
 - ☐ Mode 4 (Software triggered strobe)
 - ☐ Mode 5 ([Hardware](#) triggered strobe)

13. What are the different types of write operations used in 8253?

There are two types write operation in 8253

- ☐ Writing a control word register
- ☐ Writing a count value into a count register

The control word register accepts data from the data buffer and initialize

- ☐ Initializing the operating modes (mode 0- mode 4)
 - ☐ Selection of counters (counter 0- counter 2)
 - ☐ Choose binary /BCD counters.
 - ☐ Loading of the counter registers.

The mode control register is a write only register and the [CPU](#) cannot read its contents.

14. Give the different types of command words used in 8259A

The command words of 8259A are classified in two groups

- ☐ Initialization command words (ICWs)
 - ☐ Operation command words (OCWs)

15. Give the operation modes of 8259A?

- ☐ Fully Nest Mode
- ☐ End of Interrupt
- ☐ Automatic Rotation
- ☐ Automatic EOI mode
- ☐ Specific Rotation
- ☐ Special Mask Mode
- ☐ Edge and level Triggered Mode
- ☐ Reading 8259 Status
- ☐ Poll command
- ☐ Special Fully Nested Mode
- ☐ Buffered Mode
- ☐ Cascade Mode

16. Define scan counter?

The scan counter has two modes to scan the key matrix and refresh the display. In the encoded mode, the counter provides binary count that is to be externally decoded to provide the scan lines for keyboard and display. In the decoded scan mode, the counter internally decodes the least significant 2 bit and provides a decoded 1 out of 4 scan on SL3-SL 3. The keyboard and display both are in the same mode at a time.

17. What is the output modes used in 8279?

8279 provides two output modes for selecting the display options.

- ☐ Display scan
 - ☐ In this mode, 8279 provides 8 or 16 character- multiplexed displays those can be organized as dual 4-bit or single 8-bit display units.
 - ☐ Display Entry 8279 allows options for data entry on the displays. The display data is entered for display from the right side or from the left side.

18. What are the modes used in keyboard modes?

- ☐ Scanned Keyboard mode with 2 Key Lockout
- ☐ Scanned Keyboard with N-Key Rollover.
- ☐ Scanned Keyboard Special Error Mode.
- ☐ Scanned Matrix Mode.

19. What are the modes used in display modes?

1. Left Entry Mode

In the left entry mode, the data is entered from the left side of the display unit.

2. Right Entry Mode

In the right entry mode, the first entry to be displayed is entered on the rightmost display.

20. What is the use of modem control unit in 8251?

The modem control unit handles the modem handshake signals to coordinate the communication between the modem and the USART.

21. List the operation modes of 8255?

a) I/O Mode

☐ Mode 0- Simple Input/Output.

☐ Mode 1- Strobe Input/Output (handshake mode)

☐ Mode 2- Strobe bi-directional mode b) Bit Set/Reset Mode.

22. What is a control word?

It is a word stored in a register (control register) used to control the operation of a program digital device.

23. What is the purpose of control word written to control register in 8255?

The control words written to control register specify an I/O function for each I/O port. The bit D7 of the control word determines either the I/O functions of the BSR function.

24. What is the size of ports in 8255?

Port - A : 8- bits Port - B : 8- bits

Port -CU : 4- bits Port -CL : 4- bits

25. What is an USART?

USART stands for universal Synchronous / Asynchronous Receiver / Transmitter. It is a programmable [communication](#) interface that can [communicate](#) by using either synchronous or asynchronous serial data.

26. What is the use of 8251 chip?

8251 chip is mainly used as the asynchronous serial interface between the processor and the external equipment.

27. The 8279 is a programmable ----- interface.

Keyboard/ Display

28. List the major components of the Keyboard/ Display interface.

☐ Keyboard section

- ☐ Scan section
- ☐ Display section
- ☐ CPU interface section

29. What is Key bouncing?

Mechanical switches are used as keys in most of the keyboard. When a key is pressed the contact bounces back and forth and settles down only after a small time delay (about 20ms). Even though a key is actuated once, it will appear to have been actuated several times. This problem is called Key Bouncing.

30. What is TXD?

TXD- Transmitter Data Output

This output pin carries serial data bits along with other information like start bit, stop bits and priority bit.

31. Define HRQ?

The hold request output requests the access of the system bus. In non-cascaded 8257 systems, this is connected with HOLD pin of CPU. In cascade mode, this pin of a slave is connected with a DRQ input line of the master 8257, while that of the master is connected with HOLD input of the CPU.

32. What is RXD?

RXD- Receive Data Input

This input pin of 8251A receives a composite stream of the data to be received by 8251A.

33. What are the internal devices of a typical DAC?

The internal devices of a DAC are R/2R resistive network, an internal latch and current to voltage converting amplifier.

34. What is settling or conversion time in DAC?

The time taken by the DAC to convert a given digital data to corresponding analog signal is called conversion time.

35. What are the different types of ADC?

The different types of ADC are successive approximation ADC, counter type ADC, flash type ADC, integrator converters and voltage to frequency converters.

36. What kind of projects have you done with Raspberry Pi?

A lot of innovative projects can be done using Raspberry Pi. Like using cameras, using it like a media center and more.

You can explain about your project in detail and how your idea will benefit the world.

37. Who created Raspberry Pi?

- ☐ ODROID
- ☐ Arduino
- ☒ Raspberry Pi foundation
- ☐ DIY

The Raspberry Pi Foundation (UK) is a charity founded in 2009 to promote the study of basic computer science in schools, and is responsible for developing a single-board computer called the Raspberry Pi.

The ODROID is a series of single-board computers and tablet computers created by an open-source hardware company located in South Korea.

Arduino is an open-source electronic prototyping platform enabling users to create interactive electronic objects.

DIY is short for Do It Yourself.

38. Raspbian is the only operating system available for Raspberry Pi?

☐ True ☒ False

Raspbian is the official operating system for all models of the Raspberry Pi (RPi), but not the only one.

Raspbian is a Linux distribution based on Debian, with all packages compiled specifically for the Raspberry Pi.

Although it is a separate project, Raspberry Pi foundation creates images of Raspbian and maintains a repository containing additional software.

There are other systems available for Pi: Pidora is a remix of Fedora, Arch Linux ARM is based on Arch, Windows 10, Adafruit Occidentalis and many others...

39. Raspberry Pi usually runs from an SD card?

☒ True ☐ False

By default Raspberry Pi runs from the OS installed on SD card, but it is possible to boot Raspberry Pi also from the USB disk.

Note: Make sure you are using the Class-10 SD card or better.

40. What is magic smoke?

- ☐ What makes Electronic Circuit work
- ☒ A joke

The magic smoke is an inside joke in the electronics field.

Apparently there is a magic smoke that lives in all electronics components and that makes them work. If the magic smoke runs out of the component, it would no longer work.

41. Hardware has a tendency not to work for the smallest reasons.

☒ True ☐ False

You might have had a bad solder joint or soldered something in backward.

You may think it sounds so silly, but it's often the smallest things that trip you up.

42. Can you netboot the Pi?

☒ YES ☐ NO

Yes. The Raspberry Pi 3 can be set up to network boot without an SD card present.

Note: Earlier models could PXE/netboot with an appropriately set up SD card.

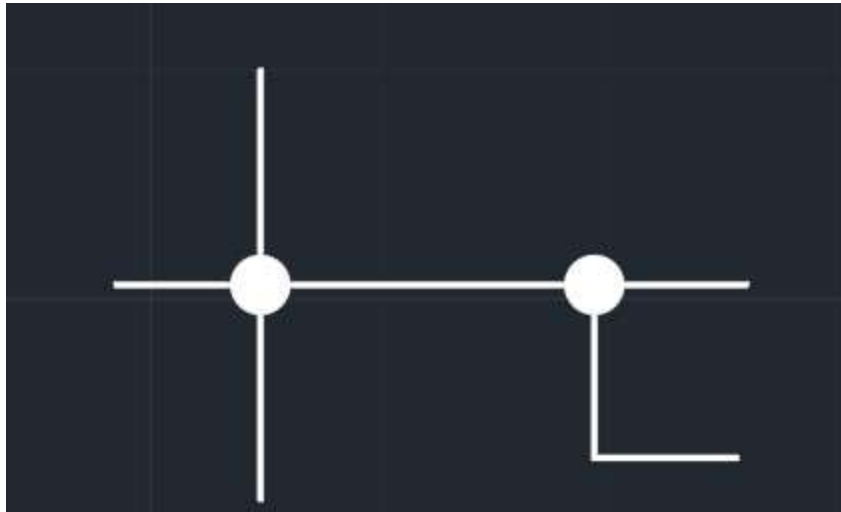
43. What is a GPIO pin?

- ☒ General Purpose Input Output pin
- ☐ Gigabyte Pin Output

GPIO stands for General Purpose Input/Output.

GPIO pin is a Raspberry Pi pin controlled by software.

44. In the image we have?



- ☒ Joined wires
- ☐ Unjoined wires

This is the joined wires schematics with the dot in the center.

Without that dot, we would have unjoined wires.

45. What will you get if you run the pinout command on Pi?

- ☐ get GPIO pin-out information
- ☐ get RPi revision number
- ☐ get model and hardware info
- ☒ All true

pinout tool is provided by the GPIO Zero Python library, which is installed by default on the Raspbian desktop image.

If you run the command pinout on Pi you can get the output like this:

```
Revision      : a02082
SoC           : BCM2837
RAM           : 1024Mb
Storage       : MicroSD
USB ports     : 4 (excluding power)
Ethernet ports : 1
Wi-fi        : True
Bluetooth     : True
Camera ports (CSI) : 1
Display ports (DSI): 1
```

J8:

3V3 (1)	(2) 5V
GPIO2 (3)	(4) 5V
GPIO3 (5)	(6) GND
GPIO4 (7)	(8) GPIO14
GND (9)	(10) GPIO15
GPIO17 (11)	(12) GPIO18
GPIO27 (13)	(14) GND
GPIO22 (15)	(16) GPIO23
3V3 (17)	(18) GPIO24
GPIO10 (19)	(20) GND
GPIO9 (21)	(22) GPIO25
GPIO11 (23)	(24) GPIO8
GND (25)	(26) GPIO7
GPIO0 (27)	(28) GPIO1
GPIO5 (29)	(30) GND
GPIO6 (31)	(32) GPIO12
GPIO13 (33)	(34) GND
GPIO19 (35)	(36) GPIO16
GPIO26 (37)	(38) GPIO20
GND (39)	(40) GPIO21

46. How do we get the power for RPi?

- ☐ Only via the motherboard
- ☒ Via motherboard or externally

Power sources don't need much representation in our schematics as the Pi is powered externally. However, some of the GPIO projects take external power as well.

Note: In the electronics world you refer to a ground or voltage source as a rail. When you see the term ground rail, for example, that would mean the ground terminal on the physical circuit.

47. What is NOOBS?

- ☒ New Out Of the Box Software
- ☐ No OBS

NOOBS is a smart OS installing tool.

NOOBS contains Raspbian OS out of the box.

NOOBS also provides a choice to install an alternative operating system that can be downloaded from the Internet.

48. Raspberry Pi foundation also provides Raspbian disk images on raspberrypi.org. From the next two images which one is meant to work from CLI only?

- ☐ Raspbian Stretch with desktop
- ☒ Raspbian Stretch Lite

Lite is a very basic Linux OS with only a command line interface (CLI).

In Lite there is no Desktop or GUI of any kind, and it does not include the educational or productivity software included with the Desktop disk image. To interact with the Lite version you will need to type commands (CLI).

For new users and people not familiar with Linux the Desktop version is recommended.

49. gEDA is?

- ☒ Software
- ☐ Hardware

The word gEDA is a conjunction of GPL and EDA.

GNU is short for General Public License. EDA is short for electronic design automation. The term gEDA refers to a set of software tools used for electronic design.

These tools are used for electrical circuit design, schematic capture, simulation, prototyping, and production.

50. Ohm's law is not important when working on Raspberry Pi projects?

☐ True ☒ False

Ohm's law is actually important when dealing with electronic circuits.

An electronic circuit is composed of individual electronic components, such as resistors, transistors, capacitors, inductors and diodes, connected by conductive wires or traces through which electric current can flow.

51. What may happen if the Pi voltage is too low?

- ☐ Pi may boot and run but crash later
- ☐ Pi may crash when using the GPIO
- ☐ Pi may freeze or work unstable
- ☒ All possible

Pi is a highly sophisticated electronic device with a very complex system on chip (SoC) on board.

Having a reliable power source is very important. The device is powered by 5V micro-USB. Exactly how much current (mA) the Raspberry Pi requires is dependent on which model you are using, and what you hook up to it.

Recommended is 2.5A (2500mA) power supply, from a reputable retailer, that will provide enough power to run your Raspberry Pi for most applications, including use of the 4 USB ports.

Very high-demand USB devices may however require the use of a powered hub. If voltage is too low, Pi may crash.

52. The on board voltage between TP1 and TP2 should be around?

- ☒ 4.75 and 5.25V
- ☐ 3.75 and 6.25V
- ☐ 12V
- ☐ It depends

If you take a look at your Pi board you will notice some small round hole contact points called TP1 and TP2.

TP1 is located under the Raspberry Pi logo and TP2 is located between the GPIO and the RCA video jack. The Pi has a voltage tolerance of 0.75 V DC or 15% tolerance. Anything outside this range is bad and you will need to investigate your power source or cable or maybe both. If your voltage reads very close to the 15% tolerance take a look at your power supply or cables.

Being very close to the 15% is just as bad as being over it; when you put the Pi under load you will go over the 15% tolerance very quickly.

53. Your multimeter shows -5V DC on TP1, TP2 probe. Is this bad?

- ☐ Very bad
- ☒ NOt bad

TP1 is where you will place the positive probe from the multimeter. TP2 is where you will place the negative probe from the multimeter.

On negative voltage readings you may think Pi went crazy. This is still a valid reading of the voltage if you ignore the negative sign. Swap the probes and check it again. It's a good idea to get into a good habit from the start.

54. What is LXDE?

- ☐ User interface
- ☐ Lightweight X11 Desktop Environment
- ☐ Free and low resource consuming software

- ☒ All true

LXDE (Lightweight X11 Desktop Environment) is a user interface which sits on top of a system called X, a client-server windowing system.

When an application starts, it requests a window in which to work from the X server program; the server also takes care of things like detecting mouse and keyboard input, and putting windows on the screen.

55. Is Python the recommended language to work with Raspberry Pi?

☒ YES ☐ NO

The name, Raspberry Pi, was the combination of the desire to create an alternative fruit-based computer (such as Apple, BlackBerry, and Apricot) and a nod to the original concept of a simple computer that can be programmed using Python (shortened to Pi).

Why has Python been selected as the language to use on the Raspberry Pi?

There are many programming languages that you can choose, from high-level graphical block programming, such as Scratch, to traditional C, right down to BASIC, and even raw machine code assembler.

Python has been selected to start when learning RPi programming because it provides a rich set of coding tools while allowing simple programs to be written without fuss.

Python is very modular with lots of additional libraries that can be imported to quickly extend the functionality.

56. You can use both Python 2 and Python 3 to program Raspberry Pi?

☒ YES ☐ NO

You can use both.

Also you have the option to convert from Python 2 to Python 3 (2to3 tool). This process is not perfect, and in some cases, you'll need to manually rewrite sections and fully retest everything.

You can write the code and libraries that will support both. The `import future` statement allows you to import the friendly methods of Python 3 and run them using Python 2.7.

57. What connection will provide the best screen resolution?

- ☒ HDMI
- ☐ Analogue (RCA socket)

For best results, use a TV or monitor that has an HDMI connection, thus allowing the best resolution display (1080p) and also digital audio output.

An alternative display method is to use the analogue composite video connection (via the RCA socket); this can also be attached to an S-Video or European SCART adapter. However, the analogue video output has a maximum resolution of 640 x 480 pixels, so it is not ideal for general use.

58. Raspberry Pi has a network connector?

- ☐ True
- ☐ False
- ☒ It depends on model
- ☐ No, you should use the USB network adapter

Network connector exists on Model B Only.

For the Model A Raspberry Pi it is possible to add a USB network adapter to add wired or even wireless networking.

59. The Raspberry Pi should work with most USB keyboards and mice available?

☒ True ☐ False

Yes, but you may encounter issues if your device draws more than 140mA, in which case a powered USB hub is recommended.

You can also use wireless mice and keyboards, which use RF Dongles. Additional configuration is required for items that use the Bluetooth dongles.

If there is a lack of power supplied by your power supply or the devices are drawing too much current, you may experience the keyboard keys appearing to stick.

60. You can run Raspberry Pi on battery?

☒ YES ☐ NO

The Raspberry Pi requires a 5V power supply, which can comfortably supply at least 700mA (1A or more recommended) with a micro USB connection.

It is possible to power the unit using portable battery packs, such as the ones suitable for powering or recharging tablets.

Note: Ensure that they can supply 5V at 700mA or over.

61. What is the proper connection?

- ☒ You should make all other connections to Pi before connecting the power
- ☐ The order of connections is not important

You should aim to make all other connections to the Raspberry Pi before connecting the power.

However, USB devices, audio, and network may be connected and removed while it is running without problems.

62. Raspberry Pi uses MBR (Master Boot Record) to bootstrap from?

☐ True ☒ False

Unlike a traditional PC, a Raspberry Pi doesn't use bootstrap code stored in the MBR. When the Raspberry Pi is powered on by default:

1. loads and runs the bootcode.bin file from the first partition /dev/mmcblk0p1 on the SD card, provided the partition has a FAT filesystem.
2. bootcode.bin in turn then loads and runs start.elf (and fixup.dat) also from the first partition.
3. start.elf then reads config.txt and sets up any GPU configuration requested.
4. start.elf then loads the kernel image and parses the command-line file from /dev/mmcblk0p2 (in most cases).
5. The Linux kernel then mounts the /dev/mmcblk0p2 partition as the root / filesystem, and continues booting the rest of the system from there.

63. You can boot Raspberry Pi from USB?

☒ True ☐ False

Usually RPi systems like Raspbian have two partitions on SD cards: BOOT and SYSTEM.

You will have to boot from the SD card but you will set the system partition in cmdline.txt file in order to boot Linux from the hard drive instead of from the SD card system partition.

Since the bootloader enumerates the USB drives asynchronously you will need to switch from MBR to GPT filesystem.

In other words you will set your hard drive to use the GUID partition table (GPT).

The latest firmware/software, however, allows booting to be set up without an SD card, for example from a USB stick or other mass storage device.

64. You installed Raspbian from an ISO image file. The size of the disk image is less than 4G and your SD card is 16GB. Can you make a system partition to use all available space on the SD card?

- ☐ Nothing this is the limitation
- ☐ Lock the SD card
- ☒ Expand the system partition on SD card
- ☐ Create another partition on SD card

A manually written image will be of a fixed size. These are made small for the small sized SD cards.

To make full use of the SD card, you will need to expand the system partition to fill the remainder of the SD card. This can be achieved using the raspi-config tool on Raspbian.

65. You use Windows to read from a Pi SD card. What file system is not supported by default in Windows?

- ☐ NTFS
- ☒ EXT3
- ☐ FAT32
- ☐ RAW

If you do need to read files from the SD card using Windows, the only non supported file system is EXT3.

RAW means the system cannot detect a valid partition. Should be ignored. NTFS and FAT32 is something that both Windows and Linux can read nowadays.

66. What is the default Raspbian desktop sharing system to connect to RPi?

- ☐ Remote Desktop
- ☒ VNC
- ☐ Teamviewer
- ☐ ARD

VNC should be installed but disabled on Raspbian by default.

VNC is just one way in which you can remotely connect to the Raspberry Pi.

ARD - Apple Remote Desktop, Teamviewer and Remote Desktop are not installed by default.

To connect from Windows to RPi you may use Remote Desktop if you install xrdp on RPi: `sudo apt-get install xrdp`

67. What will happen if you remove the SD card (system partition). Will Pi work?

- ☐ it will soon crash
- ☒ system may become unstable and opening programs may not work
- ☐ it will work and open programs without problems
- ☐ system will freeze

Pi is meant to work with the system partition set on either SD card or USB.

If we remove the system partition the system will still work since it is loaded into RAM. However, the programs that are not loaded into RAM cannot work.

Note: For Pi 2 and 3 Model B system RAM is 1GB.

68. How can you enable SSH on Raspbian?

- ☒ go to Raspberry Pi configuration
- ☐ go to Raspberry Pi recommended software and install it

Raspbian already has an SSH server installed but disabled by default. There are multiple ways to enable SSH.

From the desktop: Launch Raspberry Pi Configuration from the Preferences menu, navigate to the Interfaces tab and select Enabled next to SSH.

The same can be done from `sudo raspi-config` in a terminal window.

You could start the SSH service with `systemctl`. `sudo systemctl enable ssh` `sudo systemctl start ssh`

SSH can be enabled by placing a file named `ssh`, without any extension, onto the boot partition of the SD card. When the Pi boots, it looks for the `ssh` file. If it is found, SSH is enabled, and the file is deleted. The content of the file does not matter: it could contain text, or nothing at all.

69. How much GPU memory is recommended on Raspberry Pi?

- ☐ 16
- ☐ 64
- ☐ 128
- ☒ The best thing you can do is experiment

You can edit `/boot/config.txt` and add or edit the following line: `gpu_mem=64`

GPU memory is set in megabytes. This sets the memory split between the CPU and GPU; the CPU gets the remaining memory. Minimum value is 16; maximum value is 192, 448, or 944, depending on whether you are using a 256M, 512MB, or 1024MB Pi. The default value is 64.

70. Can I update Raspberry Pi firmware?

☒ YES ☐ NO

Raspberry Pi is divided into three sections:

1. GPU firmware (not open source),
2. ARM Linux kernel (modified)
3. user space software On Raspbian, you can update the GPU firmware using the rpi-update command.

rpi-update should be installed by default on Raspbian, but if this is not the case you run: `sudo apt-get install rpi-update`

To update the GPI firmware you run: `sudo rpi-update`

Please reboot the device to activate the new firmware.

Note: Updating the firmware may also update the kernel modules.

71. Is RPi suitable for 24/7 work?

☒ YES ☐ NO

Raspberry Pi is perfectly suitable for running continuously.

Draws very little power, can be used for a number of server tasks that imply continuous uptime.

72. What is the RPi SoC manufacturer?

- ☒ Broadcom
- ☐ Samsung
- ☐ MediaTek
- ☐ Qualcomm

All models of Raspberry Pi use Broadcom SoC's containing a Videocore4 GPU, but with various ARM CPU cores.

A System on a Chip (SoC) is a method of placing all necessary electronics for running a computer on a single chip. Instead of having an individual chip for the CPU, GPU, USB controller, RAM, northbridge, southbridge, and so on, everything is compressed down into one tidy package.

73. Is there a self-assembly kit for RPi?

☐ YES ☒ NO

It would be too expensive for us to provide kits alongside finished boards, which would mean introducing another step in manufacturing.

A kit would also be impossible to hand-solder. RPi's are assembled using special equipment robots.

74. Is there an RTC (Real Time Clock)?

☐ YES ☒ NO

There is no RTC as that would be surprisingly expensive.

One can set RTC using the GPIO pins for an interesting electronics project.

75. Can RPi add extra memory?

☐ YES ☒ NO

With SoC there are some limitations.

The RAM on the model A, A+, B, B+, and Zero is a Package on Package (POP) on top of the SoC, so it is not removable or swappable.

The RAM on the Pi 2 and 3 Model B versions is on a separate chip on the bottom of the PCB, but 1GB is the maximum RAM that the SoC used by the Pi 2 and 3 Model B versions can support.

76. RPi has an integrated Camera?

☐ YES ☒ NO

The Camera Module is a small PCB (Printed Circuit Board) that connects to the CSI-2 camera port on the Raspberry Pi using a short but extendable ribbon cable.

Current default camera module is Sony IMX219 (V2) which is an improvement over the old Omnivision OV5647 (V1). These are the only cameras that are compatible with the Raspberry Pi.

77. How many GPIO pins are there?

- ☐ 26
- ☐ 40
- ☐ depends on a model
- ☐ always 33

Depending on the model, the Raspberry Pi has either 40 or 26 dedicated GPIO pins.

In all cases, these include a UART, I2C bus, a SPI bus with two chip selects, I2S audio, 3V3, 5V, and ground. The maximum number of GPIOs can theoretically be indefinitely expanded by making use of the I2C or SPI bus.

Note: Depending on revision number the GPIO pins may be labeled differently.

78. You have an analog-to-digital converter (ADC) that you want to use with your Raspberry Pi. What would you use to connect if speed is important?

- ☒ SPI
- ☐ I2C

SPI is generally faster than I2C.

SPI allows serial transfer of data between the Raspberry Pi and peripheral devices, such as analog-to-digital converter (ADC) among other devices.

I2C is a serial protocol for two-wire interfaces to connect low-speed devices. It was invented by Philips and now it is used by almost all major IC manufacturers.

The I2C wires are known as the SDA and SCL lines; SDA stands for Serial Data, and SCL is Serial Clock.

79. Can you supply power to Raspberry Pi via GPIO 5V pin?

☒ YES ☐ NO

It is possible but you need to be careful.

When using 5V GPIO pins to add more power (usually pins #2 and #4) you are bypassing the input fuse that is on the RPi. This means you have no protection in case you make some shortcut that can damage your Pi.

You are also bypassing the TVS diode so you are basically without any protection.

80. You want to connect external electronics to your Raspberry Pi, but don't want to accidentally damage or break it.

- ☒ Do not put more than 3.3V on any GPIO pin
- ☐ Do not draw more than 3mA per output
- ☐ Do not power the Pi with more than 5V
- ☐ All good

There is no doubt about it, the Raspberry Pi is a little fragile when it comes to adding external electronics.

There are some rules of thumbs.

Do not power the Pi with more than 5V and do not put more than 3.3V on any GPIO pin.

You can get away with drawing more, but this may shorten the life of the Pi. 3mA is enough to light a red LED with a 470Ω series resistor.

Also do not poke at the GPIO connector with a screwdriver or any metal object when the Pi is powered up. Do not draw more than a total of 50mA from the 3.3V supply pins. Do not draw more than a total of 250mA from the 5V supply pins. Also do not draw more than 3mA per output.

81. How can you check your RPi revision info?

- ☐ check mounting holes
- ☐ cat /proc/device-tree/model
- ☐ cat /proc/cpuinfo
- ☒ all true ***4

Each variant of the Raspberry Pi has a different Pi Revision Number.

On Raspbian you can retrieve the Pi model as a string using either of these:

1. cat /proc/device-tree/model
2. cat /proc/cpuinfo
3. raspi-config which will prompt the revision value.

The changes among revisions include mounting holes, color of the audio socket, modifications to the power supply circuitry, different GPIO headers and varying numbers of USB ports....

The Pi 2 and Pi 3 introduced new CPUs and additional memory. Finding your Pi Revision Number is a useful technique to identify what model of Pi you have and, in some cases, where it was manufactured.

82. You have an I2C device that you want to use with your Raspberry Pi, so you want to know how to make I2C work with the Raspberry Pi. What to do?

- ☒ Use the Device Tree
- ☐ comment I2C from raspi-blacklist.conf

Adding a new device driver module to the kernel doesn't automatically result in a module being loaded - the appropriate nodes must also be added to the DT (Device Tree).

Use raspi-config to enable IC2.

Note: Editing `/etc/modprobe.d/raspi-blacklist.conf` and commenting out `#blacklist i2c-bcm2708` is no longer in use since RPi switched to the DT architecture.

83. You have an I2C device attached to your Raspberry Pi. How to check and find I2C addresses in use?

- ☒ use i2cDetect tool
- ☐ there are no addresses

i2cDetect is a useful diagnostic tool and one that is worth running the first time you use a new I2C device `i2cdetect -y 1`

It scans I2C bus for devices and outputs the addresses in use (a table with the list of detected devices).

Note: i2cDetect program can confuse your I2C bus and cause data loss.

84. Raspberry Pi has BIOS?

☐ YES ☒ NO

The Raspberry Pi uses a configuration file instead of the BIOS you would expect to find on a conventional PC.

The system configuration parameters, which would traditionally be edited and stored using a BIOS, are stored instead in an optional text file named `/boot/config.txt`

This is read by the GPU before the ARM CPU and Linux are initialised. It must therefore be located on the first (boot) partition of your SD card, alongside `bootcode.bin` and `start.elf`.

85. Whose GPIO pins respond quicker?

- ☐ Raspberry Pi
- ☒ Arduino

Even if we use the fastest C library wrapped in Python `RPi.GPIO` (`apt-get install python-rpi.gpio`) to make the IO as fast as possible Pi will perform slower than Arduino.

Pi is not really designed as a microcontroller, so the GPIO pins will not respond as quickly as an Arduino.

86. You want to run the Arduino IDE on a Raspberry Pi so that you can write and upload programs to Arduino. Is this possible?

☒ YES ☐ NO

The Arduino IDE is available for the Raspberry Pi.

Turns out you need to install arduino like this:

```
$sudo apt-get update  
$sudo apt-get install arduino
```

The Arduino IDE connects to the Raspberry Pi through its USB cable to program it.

The Arduino IDE includes a feature called the serial monitor, which allows you to both send text messages to the Arduino and see messages from the Arduino over the USB cable.

87. What is Raspberry Pi?

Raspberry Pi is a microcomputer that we can connect with any keyboard, mouse or a monitor. Raspberry Pi is able to perform all tasks like creating word documents, spreadsheet, browsing, coding, games and much more. It is a tiny Linux computer that is used by many companies for performing all computer-based activities, learning and electronic projects. The Raspberry Pi uses a 32 bit ARM processor and was first developed by Raspberry Pi Foundation. The low cost makes it more popular among young people.

88. How does the Raspberry Pi work?

The memory card slot is used for inserting an SD card that acts as the hardware of Raspberry Pi. The USB port, HDMI port, and the audio/video port help us connect with a monitor, TV or any other device. This is how Raspberry Pi is capable of working anytime and anywhere. The processor gives the correct speed for running our computer programs all the time.

89. Tell about any interesting project with Raspberry Pi.

Recently, the Astro Pi project is accelerating the use of Raspberry Pi in space stations. This is aiming to write codes in space using Raspberry Pi.

90. How is Raspberry Pi used in IoT?

Raspberry Pi can be used as a platform to develop many Internet of Things project. It is simple to use Raspberry Pi because it uses Linux OS in a small card like a computer. It is also available in

low cost. IoT, on the other hand, refers to a network of devices and software that help us to connect and exchange data. Raspberry Pi is easy to set up, so it is recommended for IoT.

91. What are the different components of a Raspberry Pi board?

The Raspberry Pi board contains a 700 or 900 MHz processor with a minimum memory provision of 128 MB. It has an additional slot for memory card too. There is a graphics set up, USB port for connecting keyboard or mouse. Raspberry Pi comes along with an audio or video output option to connect your monitor. There is an HDMI port too for connecting with TV.

92. What is the NOOBS software all about?

NOOBS or New Out Of The Box Software is used for initializing Raspberry Pi for the first time. This is the easiest way possible to install Raspberry Pi anywhere. It requires no additional access for network or any imaging software. To set up Raspberry Pi, we need the NOOBS zip file which is to be downloaded. After that we can insert an SD card, connect Monitor, keyboard or mouse and start using Raspberry Pi.

93. What are GPIO Pins used in Raspberry Pi boards?

GPIO is the acronym for General Purpose Input Output Pin. They are more like switches on the board that gives an output voltage when high and give no voltage when turned to low. This is used in the Raspberry Pi boards to make an interface between the Raspberry Pi and all the components of the board. This enables multiple interactions and makes internal properties of devices easily available on board.

94. Can Raspberry Pi be used as a server?

Yes, like we use Raspberry Pi for the desktop we can use it for the server as well. Raspberry Pi allows us to utilise it like a Web server that can help us create a simple site or even store data in the cloud. This can be accessed at any time. We can develop new templates without paying any monthly hosting fee. Raspberry Pi is thus an affordable web server too. This is very suitable for small business and for people who want to learn easy web languages.

95. What is the language used by Raspberry Pi?

Raspberry Pi uses Python as its official programming language. Python is one of the most user-friendly programming language used. It is also preferred by many companies for system development. Raspberry Pi helps us to quickly release our projects with Python. Raspberry Pi is preloaded with Python which has comprehensive syntax.

96. How is Raspberry Pi different from Arduino?

Raspberry Pi uses Linux OS and is a general purpose microcomputer. It is capable of running multiple programs at a time, while the Arduino is a simple microcomputer that is capable of running one program only.

97. Why do you think we must use the Raspberry Pi?

Raspberry Pi is ideal for creating projects. We can use it when we need computers but not advanced processing and hence save space. Raspberry Pi can be used as an alternative to a desktop or even use it as a photography tool at a low cost. There are options to create Google Home, light sensing switch and a retro gaming console. The Raspberry Pi has a wide range of uses that are essential in companies that idealize innovation.

98. Are we likely to confront any overheating problems with Raspberry Pi?

No, Raspberry Pi is free of overheating issues and does not require a fan for cooling too. It is designed in such a way that it can withstand about 100 degree Celsius. This will help us finish projects in the safest way possible.

99. How can you measure power consumption used by Raspberry Pi?

The power consumption for Raspberry Pi varies from model to model. However, it is relatively low than a complete computer system. It can be measured using a multimeter.

100. What are the generations of Raspberry Pi available?

Starting from Raspberry Pi Zero, there have been many improvements in the versions. Raspberry Pi 1, Pi 2, Pi 3 are available and A, A+, B indicate the power consumption.

101. <https://www.vskills.in/interview-questions/raspberry-pi-interview-questions>